

2.6 Derivatives and rate of change.

Definition 1: The derivative of a function f at a number a , denoted $f'(a)$, is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (1)$$

if this limit exists.

$$(1) \Leftrightarrow f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad (2)$$

(with $x = a+h$ then $h = x-a$)

Remark: $f'(a)$ is the rate of change of $y = f(x)$ with respect to x when $x = a$
(instantaneous rate of change at $x = a$)

$f'(a)$ is the slope of the tangent line to the graph of $y = f(x)$ at the point where $x = a$

The equation of a line through the point (x_1, y_1) with slope m is

$$y - y_1 = m(x - x_1) \quad (3)$$

Example 1: Determine the equation of the tangent line to the given curve at the

given $x = a$.

(a) $f(x) = x^2 + 1$ at $x = 1$

b) $f(x) = \frac{1}{x+2}$ at $x = 3$

(c) $g(x) = \sqrt{x+1}$ at $x = 0$

Example 2: Does $f'(0)$ exist if $f(x) = |x|$?

Example 3: The displacement (in meters) of a particle moving in a straight line is given by $s(t) = \frac{1}{t^2}$, (t in seconds). Find the velocity of the particle at times $t = 0$, $t = 1$ and $t = 2$.

Example 1

$$\begin{aligned} (a) \quad \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} &= \lim_{h \rightarrow 0} \frac{(1+h)^2 - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 + 2h + h^2 - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{2h + h^2}{h} \\ &= \lim_{h \rightarrow 0} (2 + h) = 2 \end{aligned}$$

$f'(1) = 2 \Rightarrow$ eq of tangent line

$$y - f(1) = 2(x - 1)$$

$$y - 2 = 2x - 2$$

$$y = 2x$$

$$\begin{aligned} (b) \quad \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} &= \lim_{h \rightarrow 0} \frac{\frac{1}{5+h} - \frac{1}{5}}{h} \quad \left(\frac{0}{0} \right) \\ &= \lim_{h \rightarrow 0} \frac{5 - 5 - h}{(5+h)5} \times \frac{1}{\cancel{h}} \\ &= \lim_{h \rightarrow 0} \frac{-\cancel{h}}{h(5+h)5} = \lim_{h \rightarrow 0} \frac{-1}{25+h} \\ &= -\frac{1}{25} \end{aligned}$$

$$f'(3) = -\frac{1}{25}$$

$\Rightarrow y - \frac{1}{5} = -\frac{1}{25}(x-3)$ is the equation of the tangent line to $y = \frac{1}{x+2}$ at $(3, \frac{1}{5})$

$$(c) \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{h+1} - 1)(\sqrt{h+1} + 1)}{h(\sqrt{h+1} + 1)}$$

$$= \lim_{h \rightarrow 0} \frac{h+1 - 1}{h(\sqrt{h+1} + 1)}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+1} + 1}$$

$$= \frac{1}{2}$$

$$g'(0) = \frac{1}{2}$$

$$\Rightarrow y - 1 = \frac{1}{2}(x - 0)$$

$$y = \frac{x}{2} + 1$$

Example 2:

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

$$\lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} -1 = -1$$

$$\lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} 1 = 1$$

$\Rightarrow \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$ does not exist.

so $f'(0)$ does not exist for $f(x) = |x|$

Example 3:

$$\lim_{h \rightarrow 0} \frac{S(a+h) - S(a)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{(a+h)^2} - \frac{1}{a^2} \right)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \times \frac{a^2 - (a+h)^2}{a^2(a+h)^2}$$

$$= \lim_{h \rightarrow 0} \frac{-2ah - h^2}{h \cdot a^2(a+h)^2}$$

$$= \lim_{h \rightarrow 0} \frac{-2a - h}{a^2(a+h)^2} = \frac{-2a}{a^4} = -\frac{2}{a^3}$$

$$S'(a) = -\frac{2}{a^3}$$

Velocity at $t=a$ is $-\frac{2}{a^3}$

—— " —— at $t=1$ is $-\frac{2}{1^3} = -2$

—— " —— $t=2$ is $-\frac{2}{2^3} = -\frac{1}{4}$